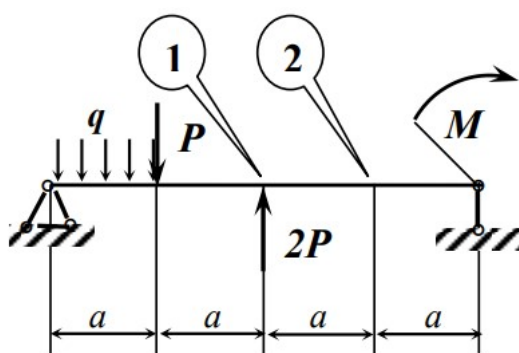


Кафедра физики прочности

Курсовая работа

по курсу

“Сопротивление материалов”



Задача № 3

Схема № 86

Группа: С23-113

Студент: Семичаев А.В.

Преподаватель: Коробов В.М.

Москва 2025

Дано

$$\begin{aligned} a_0 &:= 0.5 \text{ м} & M & & A_0 &:= 0 \text{ м} & \gamma &:= 2 \\ q_0 &:= 65000 \frac{\text{Н}}{\text{м}} & & & B_0 &:= 0.5 \text{ м} & \sigma &:= 160 \cdot 10^6 \cdot \text{Па} \\ P_0 &:= q_0 \cdot a_0 & & & C_0 &:= 1 \text{ м} & & \\ M_0 &:= q_0 \cdot a_0^2 & & & D_0 &:= 1.5 \text{ м} & & \\ & & & & E_0 &:= 2 \text{ м} & & \end{aligned}$$

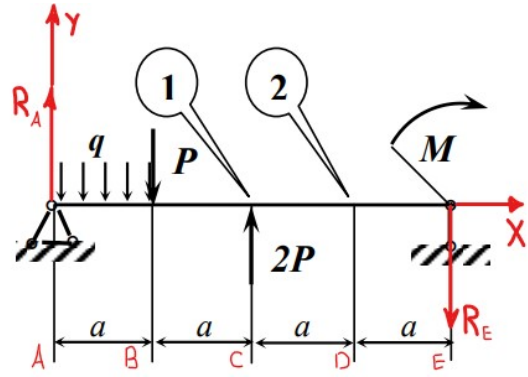
1. Определение реакций опор

1.1 Сумма всех сил вдоль оси у ноль

$$-q \cdot a - P + 2 \cdot P + R_A - R_E = 0$$

$$R_{A,2}(a, q, P, R_E) := -q \cdot a - P + 2 \cdot P + R_A - R_E \xrightarrow{\text{solve}, R_A} a \cdot q + (R_E - P)$$

$$R_{A,2}(a, q, P, R_E) \rightarrow a \cdot q + (R_E - P)$$



1.2 Сумма всех вращающих моментов ноль (рассмотрим вращение вокруг оси в точке A)

$$-q \cdot a \cdot \frac{a}{2} - P \cdot a + 2P \cdot 2a - R_E \cdot 4a - M = 0$$

$$R_{E,1}(a, q, P, M) := -q \cdot a \cdot \frac{a}{2} - P \cdot a + 2P \cdot 2a - R_E \cdot 4a - M \xrightarrow{\text{solve}, R_E} \frac{-(a^2 \cdot q) + (6 \cdot P \cdot a - 2 \cdot M)}{8 \cdot a}$$

1.3

$$R_{E,0} := R_{E,1}(a_0, q_0, P_0, M_0) \rightarrow 12187.5 \cdot \text{Н}$$

$$R_{A,0} := R_{A,2}(a_0, q_0, P_0, R_{E,0}) \rightarrow 12187.5 \cdot \text{Н}$$

2. Определение Qy и Mz

2.1

AB

$$\begin{aligned} M_{z,AB,x,2}(q, R_A, x, A) &:= R_A \cdot (x - A) - \int_0^{x-A} (t - A) \cdot q \, dt \\ M_{z,AB,x}(x) &:= M_{z,AB,x,2}(q_0, R_{A,0}, x, A_0) \rightarrow -1.0 \cdot \frac{32500.0 \cdot \text{Н} \cdot x^2}{\text{м}} + 12187.5 \cdot \text{Н} \cdot x \end{aligned}$$

BC

$$\begin{aligned} M_{z,BC,x,2}(q, R_A, x, A, B, P) &:= R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) \\ M_{z,BC,x}(x) &:= M_{z,BC,x,2}(q_0, R_{A,0}, x, A_0, B_0, P_0) \rightarrow -32500.0 \cdot \text{Н} \cdot (x - 0.25 \cdot \text{м}) + (12187.5 \cdot \text{Н} \cdot x - 32500.0 \cdot \text{Н} \cdot (x - 0.5 \cdot \text{м})) \end{aligned}$$

CD

$$\begin{aligned} M_{z,CD,x,2}(q, R_A, x, A, B, P, C) &:= R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2P \cdot (x - C) \\ M_{z,CD,x}(x) &:= M_{z,CD,x,2}(q_0, R_{A,0}, x, A_0, B_0, P_0, C_0) \rightarrow -32500.0 \cdot \text{Н} \cdot (x - 0.25 \cdot \text{м}) + (65000.0 \cdot \text{Н} \cdot (x - 1.0 \cdot \text{м}) + 12187.5 \cdot \text{Н} \cdot x - 32500.0 \cdot \text{Н} \cdot (x - 0.5 \cdot \text{м})) \end{aligned}$$

DE

$$\begin{aligned} M_{z,DE,x,2}(q, R_A, x, A, B, P, C) &:= R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2P \cdot (x - C) \\ M_{z,DE,x}(x) &:= M_{z,DE,x,2}(q_0, R_{A,0}, x, A_0, B_0, P_0, C_0) \rightarrow -32500.0 \cdot \text{Н} \cdot (x - 0.25 \cdot \text{м}) + (65000.0 \cdot \text{Н} \cdot (x - 1.0 \cdot \text{м}) + 12187.5 \cdot \text{Н} \cdot x - 32500.0 \cdot \text{Н} \cdot (x - 0.5 \cdot \text{м})) \end{aligned}$$

2.2

AB

$$\begin{aligned} Q_{y,AB,x,2}(q, x, R_A) &:= R_A - q \cdot x + Q_{y,AB,x} \xrightarrow{\text{solve}, Q_{y,AB,x}} q \cdot x - R_A \\ Q_{y,AB,x}(x) &:= Q_{y,AB,x,2}(q_0, x, R_{A,0}) \rightarrow \frac{65000.0 \cdot \text{Н} \cdot x}{\text{м}} - 12187.5 \cdot \text{Н} \end{aligned}$$

BC

$$\begin{aligned} Q_{y,BC,x,2}(q, x, R_A, P, A, B) &:= R_A - q \cdot (B - A) - P + Q_{y,BC,x} \xrightarrow{\text{solve}, Q_{y,BC,x}} (B - A) \cdot q + (P - R_A) \\ Q_{y,BC,x}(x) &:= Q_{y,BC,x,2}(q_0, x, R_{A,0}, P_0, A_0, B_0) \rightarrow 52812.5 \cdot \text{Н} \end{aligned}$$

CD

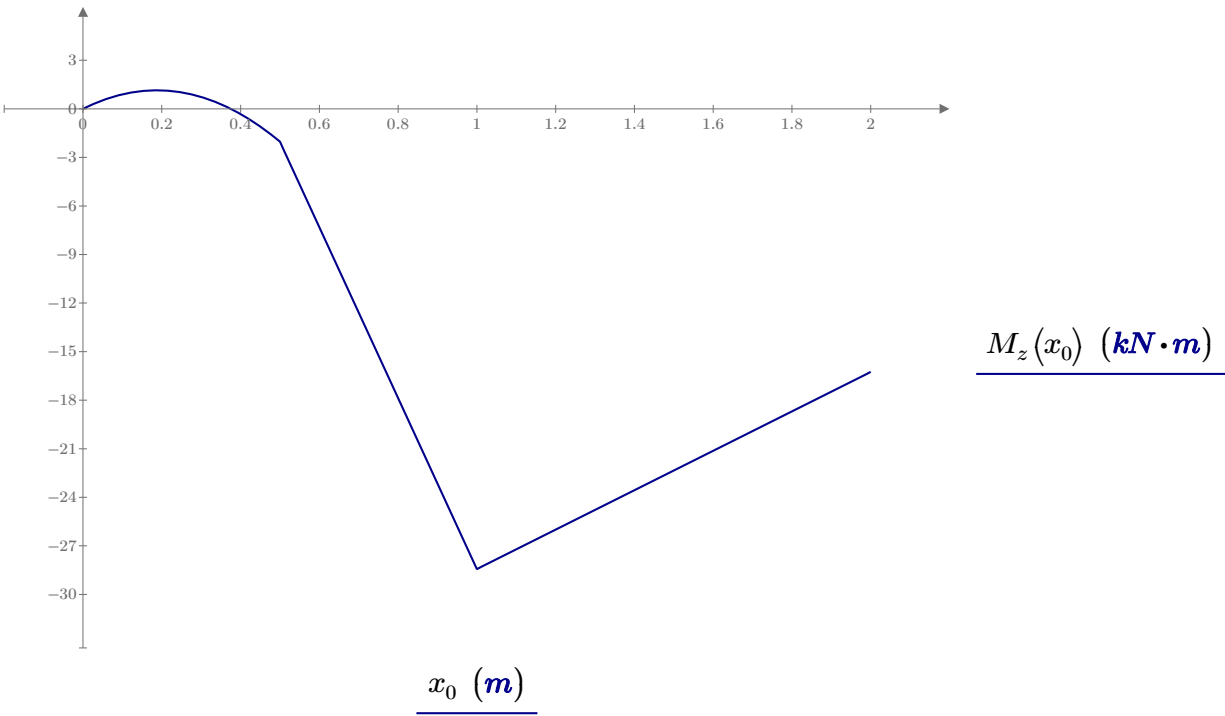
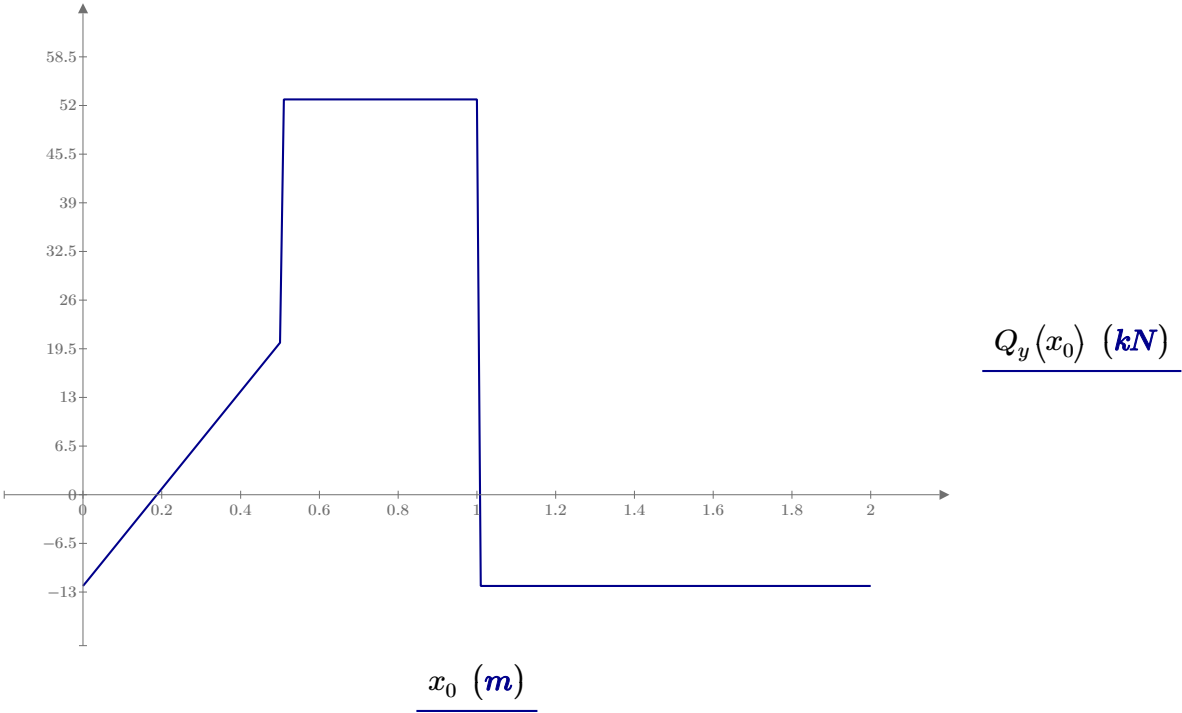
$$\begin{aligned} Q_{y,CD,x,2}(q, x, R_A, P, A, B) &:= R_A - q \cdot (B - A) - P + 2P + Q_{y,CD,x} \xrightarrow{\text{solve}, Q_{y,CD,x}} (B - A) \cdot q - (P + R_A) \\ Q_{y,CD,x}(x) &:= Q_{y,CD,x,2}(q_0, x, R_{A,0}, P_0, A_0, B_0) \rightarrow -12187.5 \cdot \text{Н} \end{aligned}$$

DE

$$\begin{aligned} Q_{y,DE,x,2}(q, x, R_A, P, A, B) &:= R_A - q \cdot (B - A) - P + 2P + Q_{y,DE,x} \xrightarrow{\text{solve}, Q_{y,DE,x}} (B - A) \cdot q - (P + R_A) \\ Q_{y,DE,x}(x) &:= Q_{y,DE,x,2}(q_0, x, R_{A,0}, P_0, A_0, B_0) \rightarrow -12187.5 \cdot \text{Н} \end{aligned}$$

$$x_0 \coloneqq A_0, 0.01 \textcolor{blue}{m}..E_0$$

$$Q_y(x) \coloneqq \left\| \begin{array}{l} \text{if } x \geq A_0 \wedge x \leq B_0 \\ \quad \left\| t \leftarrow Q_{y,AB,x}(x) \right\| \\ \text{if } x > B_0 \wedge x \leq C_0 \\ \quad \left\| t \leftarrow Q_{y,BC,x}(x) \right\| \\ \text{if } x > C_0 \wedge x \leq D_0 \\ \quad \left\| t \leftarrow Q_{y,CD,x}(x) \right\| \\ \text{if } x > D_0 \wedge x \leq E_0 \\ \quad \left\| t \leftarrow Q_{y,DE,x}(x) \right\| \end{array} \right\| \quad \left\| \begin{array}{l} M_z(x) \coloneqq \text{if } x \geq A_0 \wedge x \leq B_0 \\ \quad \left\| t \leftarrow M_{z,AB,x}(x) \right\| \\ \text{if } x > B_0 \wedge x \leq C_0 \\ \quad \left\| t \leftarrow M_{z,BC,x_0}(x) \right\| \\ \text{if } x > C_0 \wedge x \leq D_0 \\ \quad \left\| t \leftarrow M_{z,CD,x_0}(x) \right\| \\ \text{if } x > D_0 \wedge x \leq E_0 \\ \quad \left\| t \leftarrow M_{z,DE,x_0}(x) \right\| \end{array} \right\|$$



3. Подбор сечений, построение эпюр в точке E

3.1 Круглое сечение

$$Q_y(E_0) = -1.219 \cdot 10^4 \text{ Н}$$

$$M_z(E_0) = -1.625 \cdot 10^4 \text{ Н} \cdot \text{м}$$

$$M_{z_{max}} := |M_{z_{BC_{x_0}}}(C_0)| \rightarrow 28437.5 \cdot \text{Н} \cdot \text{м}$$

определение минимального диаметра:

$$W_{z_{круг}}(d) := \frac{\pi \cdot d^3}{32} \cdot \text{м}^3 \quad I_{z_{круг}}(d) := \frac{\pi \cdot d^4}{64} \cdot \text{м}^4$$

$$W_z \geq \frac{M_{z_{max}}}{\sigma} \rightarrow W_z \geq \frac{0.000177734375 \cdot \text{Н} \cdot \text{м}}{\text{Па}}$$

$$d := \sqrt[3]{\frac{M_{z_{max}} \cdot 32}{\sigma \cdot \pi}} = 0.122 \text{ м}$$

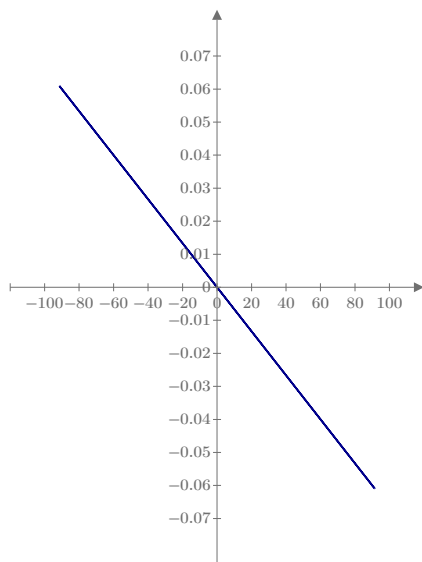
Построение эпюр

$$I_{z_{круг}}(d) := \frac{\pi \cdot d^4}{64}$$

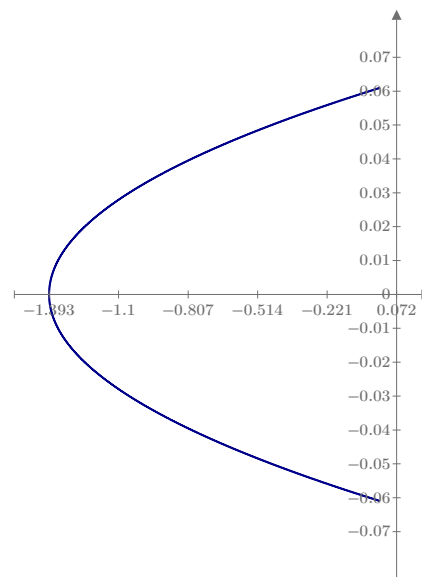
$$\sigma_{x_{круг}}(y) := \frac{M_z(E_0)}{I_{z_{круг}}(d)} \cdot y$$

$$\tau_{xy_{круг}}(y) := \frac{Q_y(E_0) \cdot \left(2 \int_0^{\sqrt{\left(\frac{d}{2}\right)^2 - (y)^2}} \frac{1}{2} \cdot \left(\left(\frac{d}{2}\right)^2 - (u)^2 - y^2 \right) du \right)}{I_{z_{круг}}(d) \cdot \left(2 \cdot \sqrt{\left(\frac{d}{2}\right)^2 - (y)^2} \right)}$$

$$y_0 := \frac{-1}{2} \cdot d, \frac{-0.9999}{2} \cdot d, \frac{1}{2} \cdot d$$



$\sigma_{x_{круг}}(y_0)$ (МПа)



$\tau_{xy_{круг}}(y_0)$ (МПа)

3.2 Прямоугольное сечение

$$Q_y(E_0) = -1.219 \cdot 10^4 \text{ N}$$
$$M_z(E_0) = -1.625 \cdot 10^4 \text{ N} \cdot \text{m}$$
$$M_{z_max} := |M_{z_BC_x=0}(C_0)| \rightarrow 28437.5 \cdot \text{N} \cdot \text{m}$$

подбор размеров сечения:

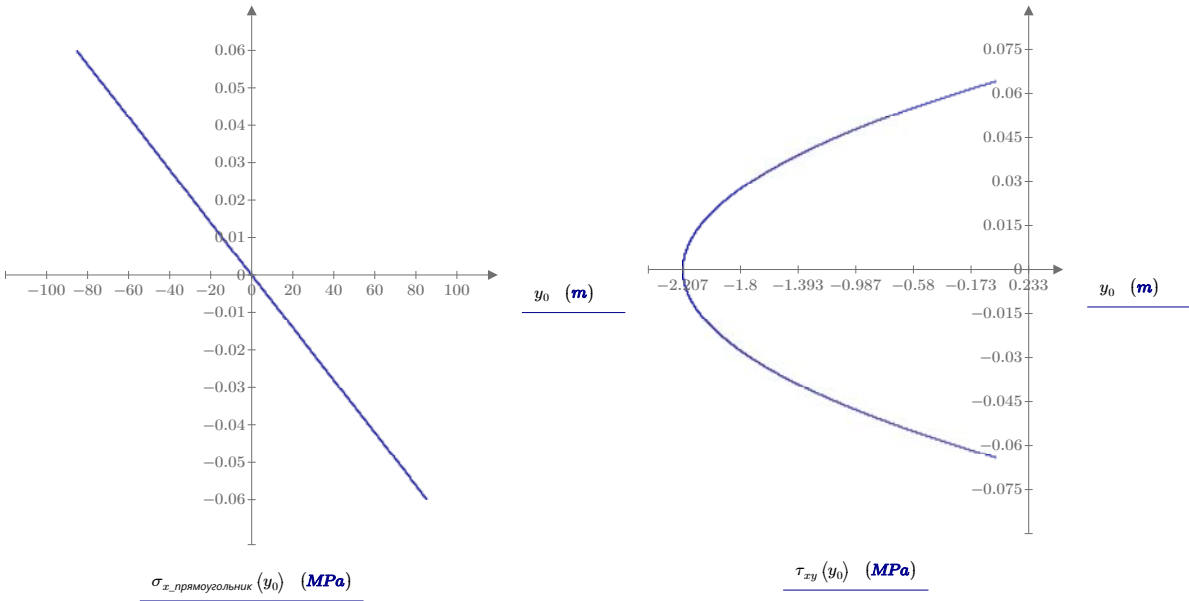
$$W_{z_брус}(b, h) = \frac{b \cdot h^2}{6} \cdot \text{m}^3$$

$$b := \sqrt[3]{\frac{6 M_{z_max}}{\sigma \cdot \gamma^2}} = 0.064 \text{ m}$$

$$b = 0.064 \text{ m}$$
$$h := \gamma \cdot b = 0.129 \text{ m}$$

Построение эпюр

$$I_{z_прямоугольник}(d, b, h) := \frac{b \cdot h^3}{12}$$
$$\sigma_{x_прямоугольник}(y) := \frac{M_z(E_0)}{I_{z_прямоугольник}(d, b, h)} \cdot y$$
$$\tau_{xy}(y) := \frac{Q_y(E_0) \cdot b \cdot \int_y^{\frac{h}{2}} v \, dv}{I_{z_прямоугольник}(d, b, h) \cdot b}$$
$$y_0 := \frac{-1}{2} \cdot h, \frac{-0.99999}{2} \cdot h .. \frac{1}{2} \cdot h$$



3.3 Двутавровая балка

$$Q_y(E_0) = -1.219 \cdot 10^4 \text{ N}$$

$$M_z(E_0) = -1.625 \cdot 10^4 \text{ N} \cdot \text{m}$$

$$M_{z_{max}} := |M_{z_{BC-x_0}}(C_0)| \rightarrow 28437.5 \cdot \text{N} \cdot \text{m}$$

Выберем двутавр 20Б1

$$h := 200.0 \cdot 10^{-3} \text{ m}$$

$$b := 100.0 \cdot 10^{-3} \text{ m}$$

$$s := 5.5 \cdot 10^{-3} \text{ m}$$

$$t := 8.0 \cdot 10^{-3} \text{ m}$$

$$h_W := 184.0 \cdot 10^{-3} \text{ m}$$

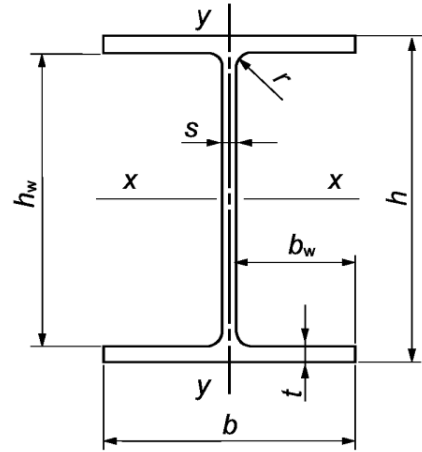
$$h_b := 47.25 \cdot 10^{-3} \text{ m}$$

$$r := 11.0 \cdot 10^{-3} \text{ m}$$

$$F_H := 27.16 \cdot 10^{-4} \cdot \text{m}^2$$

$$I_x := 1844.26 \cdot 10^{-8} \cdot \text{m}^4$$

$$I_z := I_x$$

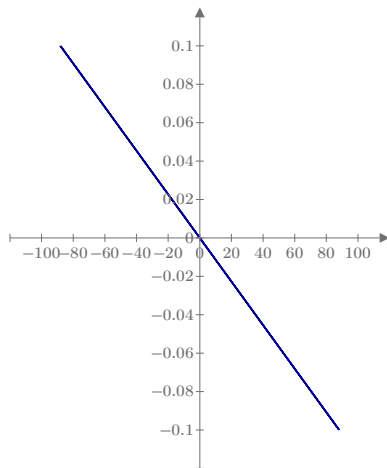


Построение эпюр

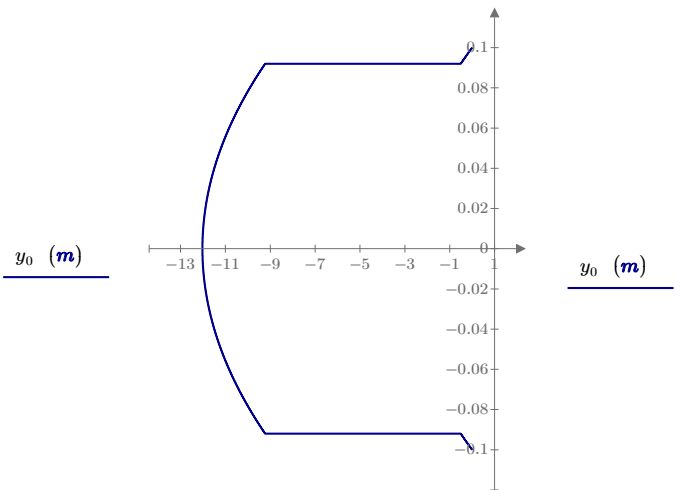
$$y_0 := \frac{-1}{2} \cdot h, \frac{-0.9999}{2} \cdot h \dots \frac{1}{2} \cdot h$$

$$\sigma_{x_{двутавр}}(y) := \frac{M_z(E_0)}{I_z} \cdot y$$

$$\tau_{xy}(y) := \begin{cases} \text{if } 0 \leq |y| < \frac{h_W}{2} \\ \tau_{xy} \leftarrow \frac{Q_y(E_0) \cdot \left(\frac{s \cdot \left(\left(\frac{h_W}{2} \right)^2 - (y)^2 \right)}{2} + \frac{b \cdot \left(\left(\frac{h}{2} \right)^2 - \left(\frac{h_W}{2} \right)^2 \right)}{2} \right)}{I_z \cdot s} \\ \text{if } \frac{h}{2} \geq |y| \geq \frac{h_W}{2} \\ \tau_{xy} \leftarrow \frac{Q_y(E_0) \cdot \left(\frac{b \cdot \left(\left(\frac{h}{2} \right)^2 - (y)^2 \right)}{2} \right)}{I_z \cdot b} \end{cases}$$



$\sigma_{x_{двутавр}}(y_0) \text{ (MPa)}$



$\tau_{xy}(y_0) \text{ (MPa)}$

Анализ напряжённого состояния в месте перехода стенки двутавра в полку

$Q_y(E_0) = -1.219 \cdot 10^4 \text{ N}$ меньше нуля, значит, касательное напряжение вверх

$$\sigma_x := \sigma_{x_{\text{двутавр}}} \left(\frac{h_W}{2} \right) = -8.106 \cdot 10^7 \text{ Pa} \quad \tau_{xy} := \tau_{xy} \left(\frac{h_W - 0.0001 \cdot m}{2} \right) = -9.231 \cdot 10^6 \text{ Pa}$$

$$\sigma_1 := \frac{\sigma_x}{2} + \sqrt{\frac{\sigma_x^2}{4} + \tau_{xy}^2} = (1.038 \cdot 10^6) \text{ Pa}$$

$$\sigma_3 := \frac{\sigma_x}{2} - \sqrt{\frac{\sigma_x^2}{4} + \tau_{xy}^2} = -8.21 \cdot 10^7 \text{ Pa}$$

$$\sigma_2 := \sigma_x$$

$$\alpha := \frac{1}{2} \cdot \text{atan} \left(\frac{2 \cdot \tau_{xy}}{\sigma_x} \right) \cdot \frac{180}{\pi} = 6.415$$

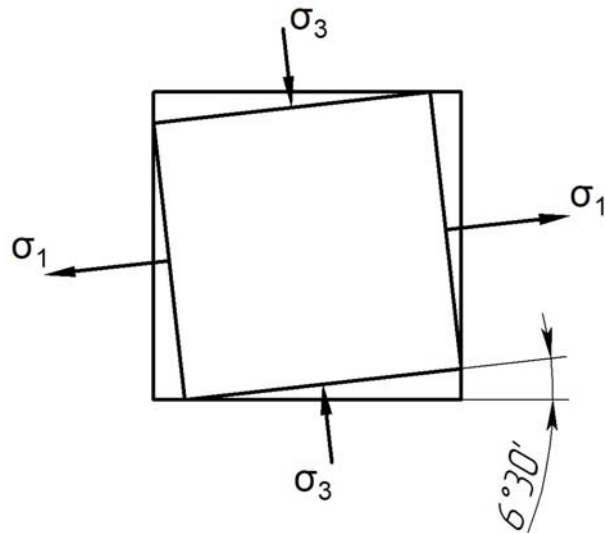
Град

$$\sigma_1 = (1.038 \cdot 10^6) \text{ Pa}$$

$$\sigma_3 = -8.21 \cdot 10^7 \text{ Pa}$$

$$\sigma_2 = -8.106 \cdot 10^7 \text{ Pa}$$

$$\alpha = 6.415^\circ$$



Интеграл Мора

$$E_{ю} := 2 \cdot 10^5 \cdot 10^6 \quad A_0 := 0 \quad \gamma := 2$$

$$a_0 := 0.5 \quad B_0 := 0.5 \quad \sigma := 160 \cdot 10^6$$

Первый пункт

$$q_0 := 65000 \quad C_0 := 1$$

$$P_0 := q_0 \cdot a_0 \quad D_0 := 1.5$$

$$M_0 := q_0 \cdot a_0^2 \quad E_0 := 2$$

$$R_{Ae} \left(R_{Ee}, P_e \right) := R_{Ae} + R_{Ee} - P_e = 0 \xrightarrow{\text{solve}, R_{Ae}} -R_{Ee} + P_e$$

$$R_{Ee} \left(A, B, E, P_e \right) := P_e \cdot (B - A) = R_{Ee} \cdot (E - A) \xrightarrow{\text{solve}, R_{Ee}} \frac{P_e \cdot B - P_e \cdot A}{E - A}$$

$$P_{e,0} := 1$$

$$R_{Ee,0} := R_{Ee} \left(A_0, B_0, E_0, P_{e,0} \right) = 0.25$$

$$R_{Ae,0} := R_{Ae} \left(R_{Ee,0}, P_{e,0} \right) = 0.75$$

AB:

$$M_{ze_AB} \left(R_{Ae}, A, x \right) := -R_{Ae} \cdot (x - A)$$

$$M_{ze_AB,0} \left(x \right) := M_{ze_AB} \left(R_{Ae,0}, A_0, x \right) \longrightarrow -0.75 \cdot x$$

BE

$$M_{ze_BE} \left(R_{Ae}, A, B, P_e, x \right) := -R_{Ae} \cdot (x - A) + P_e \cdot (x - B)$$

$$M_{ze_BE,0} \left(x \right) := M_{ze_BE} \left(R_{Ae,0}, A_0, B_0, P_{e,0}, x \right) \longrightarrow 0.25 \cdot x - 0.5$$

$$R_{E,0} := R_{E,1} \left(a_0, q_0, P_0, M_0 \right) \longrightarrow 12187.5$$

$$R_{A,0} := R_{A,2} \left(a_0, q_0, P_0, R_{E,0} \right) \longrightarrow 12187.5$$

AB

$$M_{z_AB_x,2} \left(q, R_A, x, A \right) := R_A \cdot (x - A) - \int_0^{x-A} (t - A) \cdot q \, dt$$

$$M_{z_AB_x} \left(x \right) := M_{z_AB_x,2} \left(q_0, R_{A,0}, x, A_0 \right) \longrightarrow -32500.0 \cdot x^2 + 12187.5 \cdot x$$

BC

$$M_{z_BC_x,2} \left(q, R_A, x, A, B, P \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B)$$

$$M_{z_BC_x,0} \left(x \right) := M_{z_BC_x,2} \left(q_0, R_{A,0}, x, A_0, B_0, P_0 \right) \longrightarrow -52812.5 \cdot x + 24375.0$$

CD

$$M_{z_CD_x,2} \left(q, R_A, x, A, B, P, C \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2 \cdot P \cdot (x - C)$$

$$M_{z_CD_x,0} \left(x \right) := M_{z_CD_x,2} \left(q_0, R_{A,0}, x, A_0, B_0, P_0, C_0 \right) \longrightarrow 12187.5 \cdot x - 40625.0$$

DE

$$M_{z_DE_x,2} \left(q, R_A, x, A, B, P, C \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2 \cdot P \cdot (x - C)$$

$$M_{z_DE_x,0} \left(x \right) := M_{z_DE_x,2} \left(q_0, R_{A,0}, x, A_0, B_0, P_0, C_0 \right) \longrightarrow 12187.5 \cdot x - 40625.0$$

$$d := 0.122$$

$$I_{z_крыз} \left(d \right) := \frac{\pi \cdot d^4}{64}$$

$$\int_0^{0.5} \frac{M_{ze_AB,0} \left(x \right) \cdot M_{z_AB_x} \left(x \right)}{E_{ю} \cdot I_{z_крыз} \left(d \right)} dx + \int_{0.5}^1 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_BC_x,0} \left(x \right)}{E_{ю} \cdot I_{z_крыз} \left(d \right)} dx + \int_1^{1.5} \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_CD_x,0} \left(x \right)}{E_{ю} \cdot I_{z_крыз} \left(d \right)} dx + \int_{1.5}^2 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_DE_x,0} \left(x \right)}{E_{ю} \cdot I_{z_крыз} \left(d \right)} dx \longrightarrow 0.0024321644534164055772$$

$$b := 0.064 \quad h := 0.129 \quad I_{z_прямоугольник} \left(b, h \right) := \frac{b \cdot h^3}{12} \quad I_{z_прямоугольник} \left(b, h \right) = 1.145 \cdot 10^{-5}$$

$$\int_0^{0.5} \frac{M_{ze_AB,0} \left(x \right) \cdot M_{z_AB_x} \left(x \right)}{E_{ю} \cdot I_{z_прямоугольник} \left(b, h \right)} dx + \int_{0.5}^1 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_BC_x,0} \left(x \right)}{E_{ю} \cdot I_{z_прямоугольник} \left(b, h \right)} dx + \int_1^{1.5} \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_CD_x,0} \left(x \right)}{E_{ю} \cdot I_{z_прямоугольник} \left(b, h \right)} dx + \int_{1.5}^2 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_DE_x,0} \left(x \right)}{E_{ю} \cdot I_{z_прямоугольник} \left(b, h \right)} dx \longrightarrow 0.0023101187201837341133$$

$$I_{z_двутавр} := 1844.26 \cdot 10^{-8}$$

$$\int_0^{0.5} \frac{M_{ze_AB,0} \left(x \right) \cdot M_{z_AB_x} \left(x \right)}{E_{ю} \cdot I_{z_двутавр}} dx + \int_{0.5}^1 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_BC_x,0} \left(x \right)}{E_{ю} \cdot I_{z_двутавр}} dx + \int_1^{1.5} \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_CD_x,0} \left(x \right)}{E_{ю} \cdot I_{z_двутавр}} dx + \int_{1.5}^2 \frac{M_{ze_BE,0} \left(x \right) \cdot M_{z_DE_x,0} \left(x \right)}{E_{ю} \cdot I_{z_двутавр}} dx \longrightarrow 0.0014341019004008834618$$

Второй пункт: в точке D приложим единичный момент

$$M_{e_0} := 1$$

$$R_{Ae} \left(M_e, A, E \right) := \frac{M_e}{E - A}$$

$$R_{Ae_0} := R_{Ae} \left(M_{e_0}, A_0, E_0 \right) = 0.5$$

$$R_{De} \left(M_e, A, E \right) := \frac{M_e}{E - A}$$

$$R_{De_0} := R_{De} \left(M_{e_0}, A_0, E_0 \right) = 0.5 \frac{1}{\mathbf{m}}$$

AD:

$$M_{z_{AD}} \left(x, A, R_{Ae} \right) := R_{Ae} \cdot (x - A)$$

$$M_{z_{AD}} \left(x \right) := M_{z_{AD}} \left(x, A_0, R_{Ae_0} \right) \longrightarrow 0.5 \cdot x$$

DE:

$$M_{z_{DE}} \left(x, A, R_{Ae}, M_e \right) := R_{Ae} \cdot (x - A) - M_e$$

$$M_{z_{DE}} \left(x \right) := M_{z_{DE}} \left(x, A_0, R_{Ae_0}, M_{e_0} \right) \longrightarrow 0.5 \cdot x - 1.0$$

AB

$$M_{z_{AB_{x,2}}} \left(q, R_A, x, A \right) := R_A \cdot (x - A) - \int_0^{x-A} (t - A) \cdot q \, dt$$

$$M_{z_{AB_{x,2}}} \left(x \right) := M_{z_{AB_{x,2}}} \left(q_0, R_{A_0}, x, A_0 \right) \longrightarrow -32500.0 \cdot x^2 + 12187.5 \cdot x$$

BC

$$M_{z_{BC_{x,2}}} \left(q, R_A, x, A, B, P \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B)$$

$$M_{z_{BC_{x,0}}} \left(x \right) := M_{z_{BC_{x,2}}} \left(q_0, R_{A_0}, x, A_0, B_0, P_0 \right) \longrightarrow -52812.5 \cdot x + 24375.0$$

CD

$$M_{z_{CD_{x,2}}} \left(q, R_A, x, A, B, P, C \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2 \cdot P \cdot (x - C)$$

$$M_{z_{CD_{x,0}}} \left(x \right) := M_{z_{CD_{x,2}}} \left(q_0, R_{A_0}, x, A_0, B_0, P_0, C_0 \right) \longrightarrow 12187.5 \cdot x - 40625.0$$

DE

$$M_{z_{DE_{x,2}}} \left(q, R_A, x, A, B, P, C \right) := R_A \cdot (x - A) - \left((x - A) - \frac{B - A}{2} \right) \cdot q \cdot (B - A) - P \cdot (x - B) + 2 \cdot P \cdot (x - C)$$

$$M_{z_{DE_{x,0}}} \left(x \right) := M_{z_{DE_{x,2}}} \left(q_0, R_{A_0}, x, A_0, B_0, P_0, C_0 \right) \longrightarrow 12187.5 \cdot x - 40625.0$$

$$\int_0^{0.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{AB_{x,2}}}(x)}{E_{\text{ю}} \cdot I_{z_{кр\text{уг}}}(d)} dx + \int_{0.5}^1 \frac{M_{z_{AD}}(x) \cdot M_{z_{BC_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{кр\text{уг}}}(d)} dx + \int_1^{1.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{CD_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{кр\text{уг}}}(d)} dx + \int_{1.5}^2 \frac{M_{z_{DE}}(x) \cdot M_{z_{DE_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{кр\text{уг}}}(d)} dx \longrightarrow -0.0044751825942861862621$$

$$\int_0^{0.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{AB_{x,2}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{прямоугольник}}}(b, h)} dx + \int_{0.5}^1 \frac{M_{z_{AD}}(x) \cdot M_{z_{BC_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{прямоугольник}}}(b, h)} dx + \int_1^{1.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{CD_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{прямоугольник}}}(b, h)} dx + \int_{1.5}^2 \frac{M_{z_{DE}}(x) \cdot M_{z_{DE_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{прямоугольник}}}(b, h)} dx \longrightarrow -0.0042506184451380707686$$

$$\int_0^{0.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{AB_{x,2}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{двутав}}}} dx + \int_{0.5}^1 \frac{M_{z_{AD}}(x) \cdot M_{z_{BC_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{двутав}}}} dx + \int_1^{1.5} \frac{M_{z_{AD}}(x) \cdot M_{z_{CD_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{двутав}}}} dx + \int_{1.5}^2 \frac{M_{z_{DE}}(x) \cdot M_{z_{DE_{x,0}}}(x)}{E_{\text{ю}} \cdot I_{z_{\text{двутав}}}} dx \longrightarrow -0.0026387474967376255698$$